



EDUCATION				
Program	Degree	Institution	% / CGPA	Year
Dual Degree (B.Tech + M.Tech)	Engineering Design, <i>Specialization - Robotics</i>	Indian Institute of Technology Madras	8.3/10	2024
Class XII	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	95.4%	2019
Class X	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	10/10	2017
- Awarded (among 30 out of 800+ students) the IIT Madras Young Research Fellowship based on remarkable research potential. 2021 - Secured 97.6457 Percentile in JEE Advanced and 99.1046 Percentile in JEE Mains out of over 1.5 Million aspirants across India. 2019 - Secured an All India Rank of 51 in UCEED(Undergraduate Common Entrance Examination for Design) conducted by IIT Bombay. 2019				
RESEARCH AND PROFESSIONAL EXPERIENCE				
Yaskawa Pvt. Ltd <i>Robotics Intern</i> <i>Jun 2022 – August 2022</i>	<i>Yaskawa Electric Pvt. Ltd. is an industry leader in Arc welding, Spot welding, Handling, Palletizing and Painting robots.</i> - Modeled and analyzed three-dimensional robotics palletizing algorithm and mechanics for mixed carton sizes. - Programmed and simulated arc welding and palletizing robots using MotoSim EG-VRC¹ and teach pendant. - Developed a C++ algorithm to effectively use robot's memory when sending data to the robot from computer.			
Young Research Fellow <i>IIT Madras</i> <i>Nov 2022 – Present</i>	<i>An illustrious program funded by the IITM batch of '76 to support undergraduate research; Guide - Prof. Satadal Ghosh</i> - Developing a hybrid algorithm incorporating online obstacle avoidance & path planning for unmanned vehicles - Simulated it in MATLAB environment; Drafting a research paper, to be submitted in IEEE Aerospace Journals			
Blurgs <i>Computer vision Intern</i> <i>Jun 2021 – Aug 2021</i>	<i>Employs real-time vision processing techniques to assist, automate, and augment the capabilities of drones in action</i> - Analyzed & benchmarked 7+ state of the art MOT² networks for the startup's first product for the Indian Navy - Used FairMOT to track and get the trail of multiple moving objects and Re-identify objects in real-time.			
Avishkar Hyperloop <i>Power Systems Engineer</i> <i>Oct 2020 – Jun 2021</i>	<i>The leading Asian student competition team with the vision to make Hyperloop transportation a reality in our country</i> - Designed a batter pack with high discharge rate for higher speeds while ensuring safe power supply to the pod - Extended battery pack life by 10% by reducing peak operational temperature by 10°C using an Al³ heat sink.			
Imagine Pvt. Ltd. <i>Software Dev. Intern</i> <i>Apr 2020 – Aug 2020</i>	<i>Enables collaboration across different locations using VR/AR for purposes such as industrial design, support and sales.</i> - Developed a Unity3D plugin in C# to upload any 3D model into a VR using Autodesk Forge APIs in real-time. - Built a basic Heroku app to upload and view any 3D model with 60+ file formats in the web & mobile browser - Achieved 100% reduction in staff-hours required for this specific task by automating entire process involved.			
PROJECTS				
Path Finder PyQt5 <i>ED3150 (Apr 2021)</i>	- Developed a 2D Path finder from scratch in python using state of the art A-Star & Dijkstra Algorithms in PyQt5 - Designed an interactive UI for visualization using QT Designer; Documented the project for future use.			
Human Powered Segway <i>(Apr 2021 – Jun 2021)</i>	- Lead a 6-member team involved in ideation of a novel mechanism of a fully mechanical alternative of a Segway - Improved efficiency by 50% by replacing conventional circular-gear pedaling system with rectilinear pedaling. - Carried out kinematic analysis to determine 6+ parameters using MATLAB and SIMULINK to improve efficiency.			
CFD⁴ Project <i>(Mar 2021 - Apr 2021)</i>	- Designed an aero foil for the Front wing of a Formula Student car for maximum Downforce/Drag ratio. - Performed CFD⁴ analysis to determine maximum possible ratio and the optimal setup for better efficiency.			
Robotic Arm <i>(Apr 2021 – Jun 2021)</i>	- Designed a small robotic arm with 2 degree of freedom for identifying and lifting objects using electromagnets - Programmed using Arduino mega and used proximity sensors for detection and LED light for sensing color			
RELEVANT COURSES AND SKILLS				
Courses	Deep Learning*, Field and Service robotics, Control Systems, Design of Mechanical Systems, Operation Research, Fundamentals of Automotive System, Mechatronics System Design. Mechanics and Control of Serial Robots*			
MOOCs^[5]	Algorithmic Toolbox*, Introduction to Reinforcement Learning*, Game Development, Introduction to Tensorflow			
Skills	Languages: Python, C, C++, C#	Tools: VS Code, MATLAB, LaTeX, MS-Office, Unity3D, Fusion360, Linux		
	Frameworks: Tensorflow, Pytorch	Libraries: Pandas, NumPy, Scikit Learn, Matplotlib, OpenCV, PyQt5, OpenGL		
POSITION OF RESPONSIBILITIES				
Department Core, Placement & Internship Cell 2022	Heading & mentoring a 3-tier team in charge of department placement and internship process 1200+ students. - Analyzed past 3 years' list of contacts to eliminate redundant calling in order to increase company conversion. Planning & executing the logistic framework for the first ever shift to hybrid mode of the recruitment process.			
Junior Consultant <i>ShARE 2021</i>	- Worked in the 'Mobility' Module, underwent training on focused application of problem-solving in consulting. - Part of a global team of 700 students across 15 nations, enabling firms to transition into 'Do Well, Do Good'			
Coding and Logic Events <i>Shaastra⁴ 2021</i>	- Handled a total footfall of 300+ teams and crafted a total of 50+ logical questions for events in Shaastra'21. Ideated and executed 2 brand new events, Mystery Rooms and E-Contest with 100+ teams participating.			
EXTRA CURRICULAR ACTIVITIES				
Drawing	Won 5+ state level competitions and 10+ District level competitions in drawing (Oil Pastels and Pencils)			
Teaching Assistant	Assisted professors and 20+ students in coursework and grading in the courses GN1102⁶ & GN6106⁷			
Fusion360 API	Developed a python script to automatically create models in Fusion360 just by specifying dimensions.			
Game Dev. / Gaming	- Love to try out various tools and features in Unity3D; Reached a rank of Silver 2 in Valorant by Riot Games. - Solely developed a complete 3D Game in Unity3D with Obstacles, Enemies, Points, UI and Visual Effects.			
Sports	Among the 20 out of 100 students selected for freshman Volleyball training; Part of the Hostel Volleyball team			

[1] Motoman Simulator Enhanced Graphics Virtual Robot Controller; [2] Multi Object Tracking; [3] Aluminum; [4] Computational Fluid Dynamics; [5] Massive Open Online Courses [6]-Lifefskills; [7]-Happiness, Habits and Success ** - Upcoming courses; * - Ongoing courses